Astrodynamics 2

2020.09.17

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Hohmann Transfer Problem

- Hohmann Transfer Problem
  - Prove that the Hohmann transfer, a larger R($= \frac{r_{final}}{r_{initial}}$) does not necessarily imply a larger Δv.

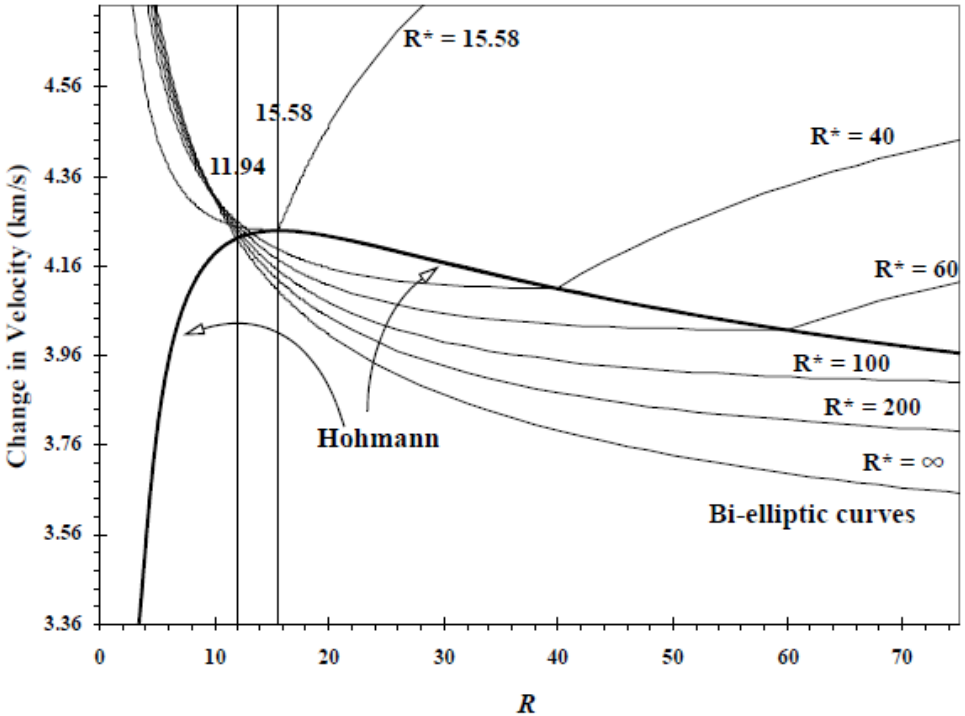


Figure 6-6. Hohmann Transfer—Comparison of Change in Velocity. The change in velocity is shown when using Hohmann and bi-elliptic orbit changes. Be aware of the different notations for the ratios (final / initial) (*R* for Hohmann and *R** for bi-elliptic). Also, for all cases, $v_{initial}$ = 1 ER/TU = 7.905 km/s.

<Hohmann Transfer-Comparison of Change in Velocity>

Hohmann Transfer Problem

- Hohmann Transfer Case

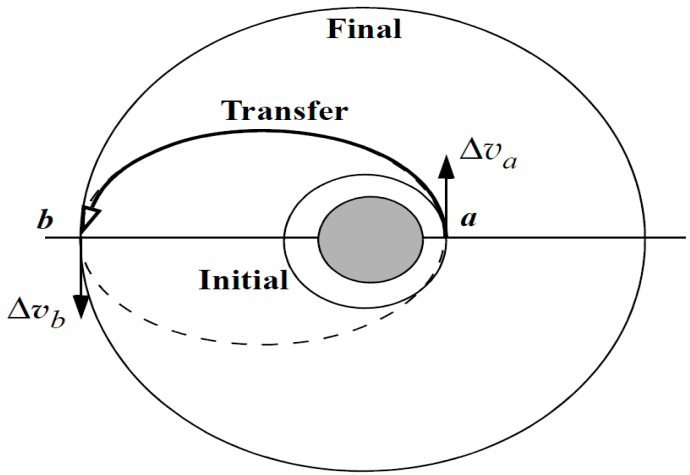
Following visa-viva equation.

$$\Delta v_a = \sqrt{\frac{2\mu}{r_{initial}} - \frac{\mu}{a_{trans}}} - \sqrt{\frac{\mu}{r_{initial}}}$$

$$= \sqrt{\frac{2\mu}{r_{initial}}\left(\frac{r_{final}}{r_{initial}+r_{final}}\right)} - \sqrt{\frac{\mu}{r_{initial}}}$$

$$\Delta v_b = \sqrt{\frac{\mu}{r_{fianl}}} - \sqrt{\frac{2\mu}{r_{final}} - \frac{\mu}{a_{trans}}}$$

$$= \sqrt{\frac{\mu}{r_{fianl}}} - \sqrt{\frac{2\mu}{r_{final}}\left(\frac{r_{initial}}{r_{initial}+r_{final}}\right)}$$



<Hohmann Transfer>

$$where\ a_{trans} = \frac{r_{initial}+r_{final}}{2}$$

$$\Delta v_{Hohmann} = |\Delta v_a| + |\Delta v_b|$$

$$= \sqrt{\frac{2\mu}{r_{initial}}\left(\frac{r_{final}}{r_{initial}+r_{final}}\right)} - \sqrt{\frac{\mu}{r_{initial}}} + \sqrt{\frac{\mu}{r_{fianl}}} - \sqrt{\frac{2\mu}{r_{final}}\left(\frac{r_{initial}}{r_{initial}+r_{final}}\right)}$$

Hohmann Transfer Problem

- Hohmann Transfer Case
  - R parameter, Function define

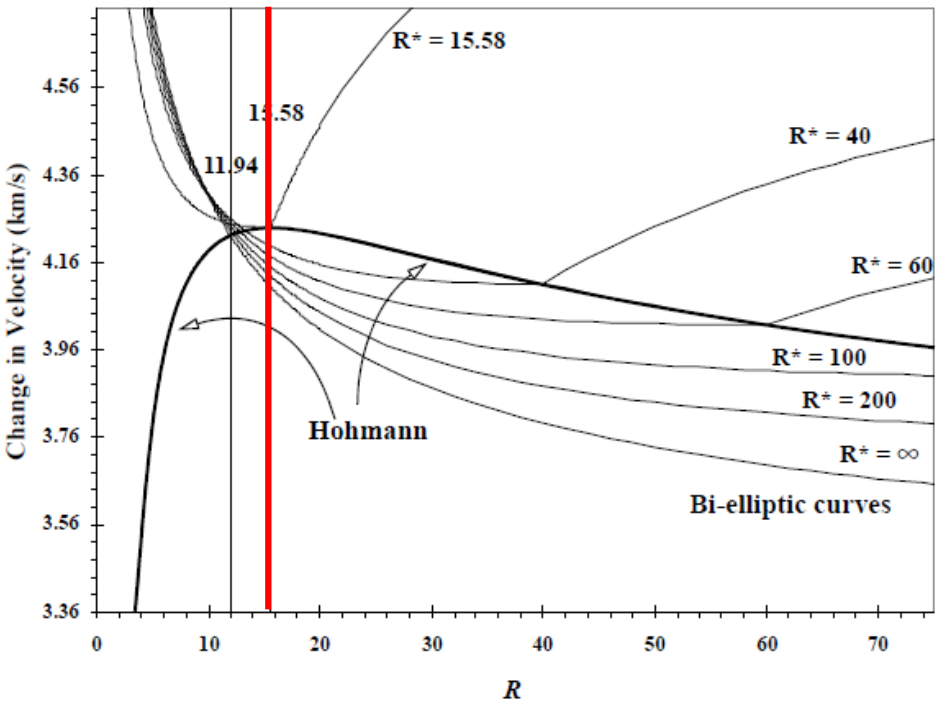
$$\frac{\Delta v_{Hohmann}}{v_{initial}} = \left(1 - \frac{1}{R}\right)\sqrt{\frac{2R}{1+R}} + \sqrt{\frac{1}{R}} - 1 \qquad where\ R = \frac{r_{final}}{r_{initial}}$$

  - Find peak R

$$\frac{\partial}{\partial R}\left(\frac{\Delta v_{Hohmann}}{v_{initial}}\right) = 0$$

$$\rightarrow\ R^3 - 15R^2 - 9R - 1 = 0$$

$$R = 15.58176$$



<Hohmann Transfer Problem>

Hohmann Transfer Problem

- Bi-elliptic Case

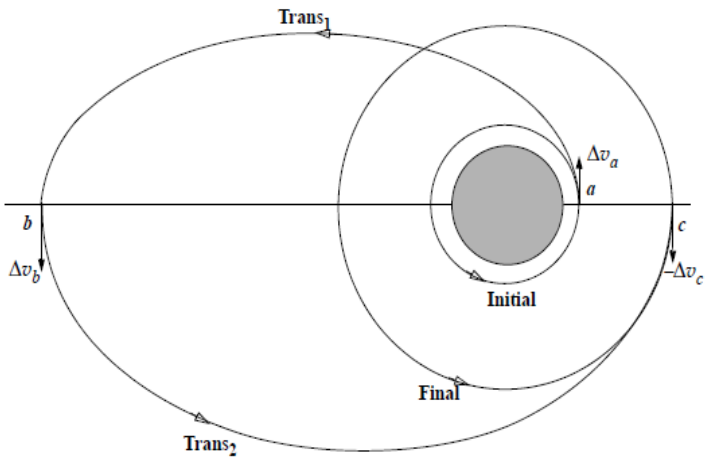
Following visa-viva equation.

Set $\boldsymbol{r_b} = \infty$

$$\Delta \mathrm{v_a} = \sqrt{\frac{2\mu}{r_{initial}} - \frac{\mu}{\infty}} - \sqrt{\frac{\mu}{r_{initial}}} = \left(\sqrt{2} - 1\right) v_{initial}$$

$$\Delta \mathrm{v_b} = 0$$

$$\Delta \mathrm{v_c} = \sqrt{\frac{2\mu}{r_{final}} - \frac{\mu}{\infty}} - \sqrt{\frac{\mu}{r_{final}}} = \left(\sqrt{2} - 1\right) v_{final}$$



<Bi-elliptic Transfer>

$$\Delta \mathrm{v}_\infty = |\Delta v_a| + |\Delta v_b| + |\Delta v_c|$$

$$= \left(\sqrt{2} - 1\right) v_{initial} + \left(\sqrt{2} - 1\right) v_{final}$$

Hohmann Transfer Problem

- Bi-elliptic Case
  - R parameter, Function define

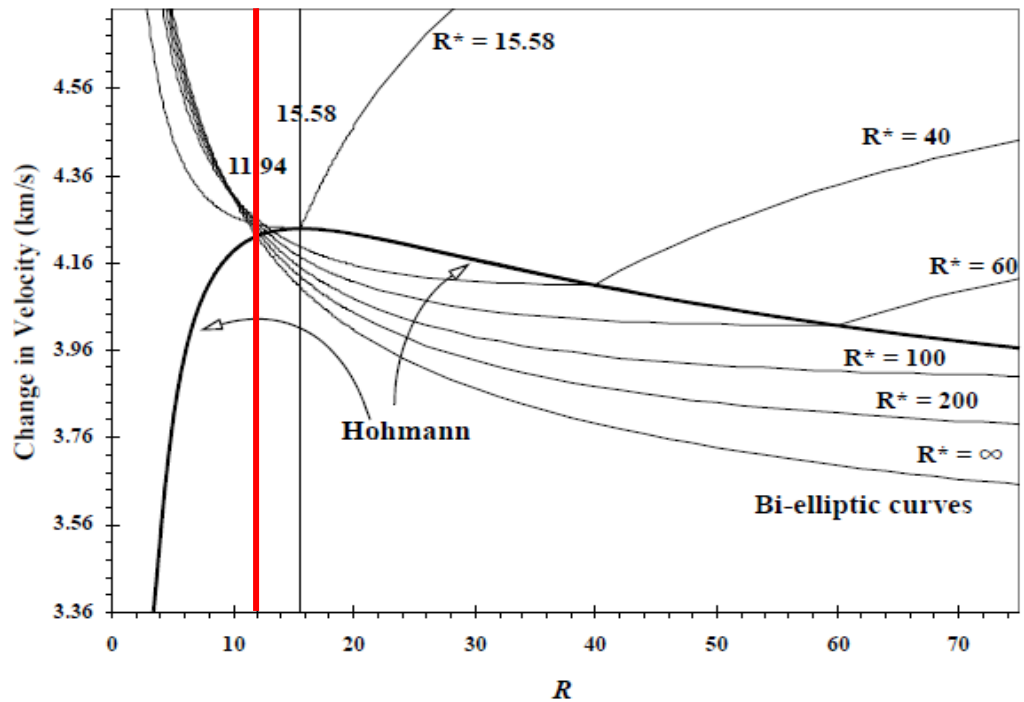
$$\frac{\Delta v_\infty}{v_{initial}} = (\sqrt{2}-1)\left(1+\sqrt{\frac{1}{R}}\right) \qquad where\ R = \frac{r_{final}}{r_{initial}}$$

  - Find $\frac{\Delta v_\infty}{v_{initial}} = \frac{\Delta v_{Hohmann}}{v_{initial}}$

$$(\sqrt{2}-1)\left(1+\sqrt{\frac{1}{R}}\right)$$

$$= \left(1-\frac{1}{R}\right)\sqrt{\frac{2R}{1+R}} + \sqrt{\frac{1}{R}} - 1$$

$$R = 11.93876$$



<Bi-elliptic Problem>

Thank you for listening